

A scenario analysis of potential long-term impacts of COVID-19 on the Tunisian electricity sector

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ABSTRACT

The COVID-19 pandemic has affected the economy of Tunisia, like that of many other countries. With electricity consumption dropping, consumption patterns changing, international fuel prices oscillating and uncertainty raging, the pandemic has affected not least the planning of investments in electricity supply. Although the government seems unlikely to revise the decarbonisation targets downwards, questions arise on whether the investments planned before the pandemic are still relevant in the changed global landscape and what effects they may have on the country's economy.

In this study, we analyse post-pandemic scenarios for the electricity supply system of Tunisia with an energy-economy modelling framework, soft-linking the energy modelling tool OSeMOSYS and an open source Input-Output model. We extract insights on the cost-competitiveness of different – previously planned and new – electricity supply solutions and their impacts on job creation and loss in the entire economy. We find that renewable solutions based on solar photovoltaic remain highly competitive with gas-fired generation under different projections of gas prices and that several low-carbon and energy efficiency solutions have high potential for job creation. We also find that more ambitious investments in renewables and energy efficiency are needed to take Tunisia towards the path of deep decarbonisation.

1. Introduction

The COVID-19 pandemic has had deep effects on the global and national economies, including high volatility of international fossil fuel prices and a sharp reduction of energy demands. Some of these effects may last beyond the time frame of new waves of spread of the virus, due to a slow recovery of national economies. The Organisation for Economic Co-operation and Development, OECD [1], shows that international oil prices have fallen and gone negative in some instances. World Bank projections [2] show they may be still far from recovering years from now (trend-wise and not considering supply crises). In Tunisia, the ministry in charge of energy matters shows that, under the pandemic, the demand for electricity has sharply decreased, mostly due to reduced industrial and service activity [3]. Gas prices, indexed to international

oil prices, followed the downward trend with a delay of nine months [4]. Oil and gas prices are expected to recover as demands picks up again after the pandemic. Further, sharp changes in supply quantities may affect the prices greatly (as happened in Europe in the summers during the pandemic). However, as long as they could be, shocks like the one created by COVID 19 could happen again.

In Tunisia, as elsewhere, there is high uncertainty over the economic recovery and the impacts the pandemic may have on the energy sector in the long term. The Government's ambitions for the decarbonisation of the energy system are stated in the Intended Nationally Determined Contribution (INDC) submitted to the UNFCCC [5] and the Nationally Appropriate Mitigation Actions (NAMA) [6] and set in motion in the Tunisian Solar Plan (PST) [7]. The main objective in the INDC is a 41% conditional (13% unconditional) reduction of carbon intensity in the economy by 2030 compared to 2010. Specific targets for the energy

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Abbreviations

ANME	National Agency for Energy Management
CCGT	Combined Cycle Gas Turbine
CSP	Concentrating Solar Power
FDI	Foreign Direct Investment
INDC	Intended Nationally Determined Contribution
IO	Input-Output
MENA	Middle East and North Africa
NAMA	Nationally Appropriate Mitigation Actions
OCGT	Open Cycle Gas Turbine
OSeMOSYS	Open Source energy Modelling System
PST	Tunisian Solar Plan (Plan Solaire Tunisien)
PV	Photovoltaic
STEG	Tunisian Company of Electricity and Gas
UNFCCC	United Nations Framework Convention on Climate Change

sector are 30% reduction of the primary energy demand by 2030 compared to a baseline and 30% share of renewable energy in electricity supply by 2030. This last target is a pillar of the PST and turned into targets of installed capacity of technologies (around 1.7 GW of wind, 1.5 of solar PV and 0.45 of concentrating solar power). These targets have not changed in the wake of the pandemic and there seems to be no intention to revise them downwards. However, the long lock-down and the uncertainty on the future may delay the approval of capacity expansion plans and therefore the achievement of policy objectives. Furthermore, the decrease in gas prices, if persisting, may affect the cost competitiveness of renewables and raise questions about the extent of planned investments for the latter.

At the same time, this crisis may open the ground to reflections on questions of security of supply, especially given the uncertainties linked to future gas prices. This could favour renewables and even lead to an increase in ambitions on their penetration.

At the level of the national economy, the reduction of oil and gas prices may have decreased the share of national budget spent on imports and allowed a re-allocation on to other sectors, like health. Furthermore, it may bear the same positive effects that it seems to bear in other OECD countries with regards to purchasing power of consumers and producer prices of goods and services. However, the relation between fall of oil prices and national economy is made complex by several factors, not least the impacts on availability of Foreign Direct Investment (FDI). Significant part of the FDI in the MENA region comes from oil exporting countries, which have suffered a fall in revenues due to dropping oil prices [8]. Therefore, overall, the short-to medium-term economic impact of lower oil and gas prices on the Tunisian economy may be negative. Nevertheless, this potential negative impact may be counter-balanced in the longer range through economy-wide gains in employment related to new energy infrastructure investments, if these bear high employment potential and significant parts of the infrastructure production chains are located in Tunisia.

While predicting the economy-wide impacts of the pandemic under such uncertainties is hard, quantitative models may help the national government and utilities to understand how the cost, competitiveness and economy-wide impacts of various policy and investment choices may change under different potential post-COVID-19 futures.

1.1. Scope

In this case study, we analyse potential long-term impacts of the COVID-19 pandemic in Tunisia with an energy-economy model. We provide insights on three main aspects:

- Whether and how the cost-competitiveness of different technological solutions for meeting future electricity demand may shift, in light of changes of demands and international fuel prices due to the pandemic and in light of potential changes in policy and investment priorities;
- Where the post-pandemic evolution of the electricity sector may take Tunisia, in the path to decarbonisation;
- What implications different investment choices in the electricity supply sector may bear to job creation and loss across the entire Tunisian economy.

We extract the insights from a range of scenarios, developed in collaboration with a consortium of institutions involved in energy research and decision making in Tunisia.

The rest of the paper succinctly presents the methodology and scenarios (section 2), key results (section 3) and policy-relevant insights (section 4). Further details on the methodology and the literature background to it are presented in MethodsX papers linked to the present work.

2. Methodology and scenarios

The analysis is carried out with an energy-economy modelling framework, that soft-links the Open Source energy Modelling System – OSeMOSYS [9] - (for the energy part) and an in-house accessible Input-Output (IO) model (for the economy part). The methodology is based on, and extends, work by Howells and Laitner [10] and by Winkler et al. [11]. The version of OSeMOSYS used for the study is the one available through the v1.10 installation of the interface MoManI, available on the tool's website [12]. Key numerical assumptions are annexed to this paper.

2.1. Input output model

The IO model is built starting from the latest Input Output Table of Tunisia of 2015 publicly available at the time of submission [13]. By means of the Leontief Inverse matrix, it calculates the multipliers, which summarise economy-wide impacts of changes in spending of one sector. In parallel, the overall economic flows related to installation and operation of every electricity supply technology are calculated. This calculation is based upon techno-economic assumptions (capital and O&M costs, useful life, efficiency, debt, equity, rates of interest, fuel prices, local content, etc.) differing for each technology and harmonised with those of the electricity system model. By means of clarity, the local content represents the ratio of money flow (related to the value chain of a technology) that is paid locally and therefore circulates within the Tunisian economy. Each of the resulting economic flows is then multiplied by the multiplier of the relative sector, to have an estimation of its economy-wide impact. Finally, such monetary flow is turned into a number of jobs created and lost per technology type, by means of assumptions on an average wage and a wage share of value added. The resulting number provides an indication of economy-wide job loss and creation coefficients associated with investment in electricity supply options and energy efficiency measures, per unit of electricity generated or saved: a job creation coefficient of 0.32 for a certain power plant, for example, suggests that 1 PJ of activity of the power plant in one year leads to the creation of 320 jobs; Similarly, a job loss coefficient of 0.58 indicates that 1 PJ of annual activity is causing a loss of 580 jobs.

All the assumptions and steps of this calculation are detailed in the MethodsX papers linked to this work. The coefficients, as computed by the IO model, are summarised in Table 1. The job creation and job loss coefficients associated with each technology are calculated separately in the IO model and they are here presented separately to show the extent of each. For the purposes of the electricity system model, they could be summed into one net job creation coefficient. Notice that some of the coefficients vary with time, due to assumptions on the decrease in

Table 1

Estimates of job-gains and job-losses per energy system element, in the base year of the modelling period. A downward facing arrow indicates a decrease of the coefficients over the years. All other coefficients are constant.

Technology	Job creation coefficient (1000 jobs/PJ - year)	Job loss coefficient (1000 jobs/PJ - year)
450 MW new combined cycle	0.32	−0.58
300 MW new gas turbine	0.40	−0.82
Onshore wind	0.67	−0.51
Offshore wind	1.16	−0.87
Wind autoproduction	0.67	−0.51
Solar PV utility scale	0.64↓	−0.46↓
CSP (Concentrating Solar Power)	1.30 ↓	−1.02↓
Solar autoproduction	0.71↓	−0.51↓
Energy Efficiency – Residential	0.41	−0.28
Energy Efficiency – Commercial	0.37	−0.21
Energy Efficiency - Industrial	0.52	−0.24

*The term Autoproduction refers to private installations aimed at self-consumption. This is detailed later.

capital, fixed O&M and variable O&M costs for the least mature technologies.

2.2. Electricity system model

The electricity system model in OSeMOSYS determines the least cost evolution of the electricity supply mix of Tunisia and dynamically computes a rough estimate of overall job gains and losses associated with electricity infrastructure investments, using the coefficients calculated in the IO model. OSeMOSYS is a long-term, dynamic and bottom-up (i.e. technologically detailed) energy system optimisation framework, in this instance used in a deterministic fashion.

The structure, development and testing of the framework are discussed in the MethodsX papers linked to this publication. The framework is developed based on the groundwork by Dhakouani et al. [14] and used locally, for analyses related to the development of the Tunisian energy system [15]. Compared to Dhakouani et al. our work enhances the representation of job losses and gains related to energy investments.

The time domain of the study covers every year from 2015 (for calibration with latest available IO data) to 2040. That is, the least-cost configuration of the electricity system is calculated in 1-year time steps. For the electricity system optimisation, each year is split in 108 time steps (7 seasons * 2 day types (weekday and weekend) * 8 parts in one day), to ensure a minimum representation of the time variability of loads and variable renewables.

The current and potential electricity supply options are represented in detail in the model and synthesised in Fig. 1. The scheme is an aggregated representation of the electricity supply system of Tunisia, from the primary energy supply (left) to the final demand (right). The rectangular boxes are current or potential electricity supply technologies/processes, whereas the coloured lines are energy vectors flowing between them. The black lines are calculated numbers of CO₂ emissions and jobs lost or created, function of the results of the energy system optimisation. The model includes the existing power supply capacity at the time of writing, which consists of gas power plants for the major part (around 4600 MW), 245 MW of wind power, 30 MW of solar photovoltaic and 62 MW of hydro. The capacity investments up to 2025 reflect committed investments by the state owned Tunisian Company of Electricity and Gas (STEG). After 2025, the allowed capacity expansions include natural gas-fired combined and open cycles, wind turbines and solar PV. As no significant Concentrating Solar Power (CSP) and biomass investments are on the horizon, only limited, fixed investments are

allowed for these (100 MW and 450 MW for wind and CSP by 2030, respectively). Also private wind and solar installations for self-consumption are included (and named ‘autoproduction’, out of a term commonly used by the power utility). Their capacity is represented as aggregated and its value is fixed throughout the time domain of the model, according to projections elaborated by STEG.

2.3. Key assumptions

The techno-economic assumptions relative to each technology are the same in the electricity system model and the IO model and are summarised in Table 2. The assumptions were collected in collaboration with the local consortium and are mostly based on data from the energy utility STEG and the National Agency for Energy Management (ANME).

The whole database populating the model is annexed to this paper and it is structured so as to help the user reconstruct the model we created (with the disclaimer that some data is aggregated due to confidentiality constraints). The database is described in detail in the MethodsX papers linked to this publication.

Besides the technology-specific assumptions, key inputs to the model (and determining its outputs) are the electricity demand, fuel prices and policies. The baseline electricity demand is based on projections by STEG. The demand projection is created bottom-up combining sectoral development data and the significant growth relies upon assumptions on fast economic growth (between 3% and 5% annually), population growth and increased household use of electricity appliances. It also takes into account increased cooling due to rising average temperatures during the summer months. The demand is modified to take into account the effects of the COVID-19 pandemic: it decreased by 6.3% in 2020 compared to 2019 (according to STEG), but it is projected to fully recover to pre-pandemic levels already in 2021 and it follows the baseline trend thereafter. The intra-annual profile of the demand is based on half-hourly values of national consumption for 2017. It is kept constant throughout the modelling period, except for the years of the pandemic. In 2020 and 2021, variations in the base load, day peak and night peak are derived from a qualitative analysis by Nouicer and Dhakouani [4] (referring to March/April 2020) and applied as percentage variations to the related time steps in the model. After 2021, the variations are linearly reduced to 0 in 2025. The projections of natural gas prices are also provided by STEG. They are assumed to fall on average by 35% within 2020 and to recover thereafter, reaching and overshooting pre-pandemic projections by 2027. Whether the recovery actually happens faster depends on many factors. Nonetheless, this scenario aims to be indicative of any potential long-term shock in gas prices. The evolution of electricity demand and gas prices, major drivers of the model results, is presented in Table 3.

In terms of policies, the targets of the Tunisian Solar Plan are introduced as minimum total (existing and cumulatively installed) capacity of wind turbines (1755 MW), of solar photovoltaic (1510 MW) and of Concentrating Solar Power (450 MW) by 2030. Also the government's targets of decrease in energy intensity of the economy are introduced, with an assumption by STEG of 0.8% annual reduction of electricity demand as a percentage of the demand in 2015 (compared to a baseline).

2.4. Modelled scenarios

We look at four fictional but realistic scenarios, including direct short-term consequences of the COVID-19 pandemic, long-term projections of these consequences in the future and policy choices that may derive indirectly from reflections made after the pandemic:

- **Constant ambition.** It represents a situation where current policies are maintained, in the aftermath of the pandemic. The energy efficiency and renewable share ambitions of the government as per the

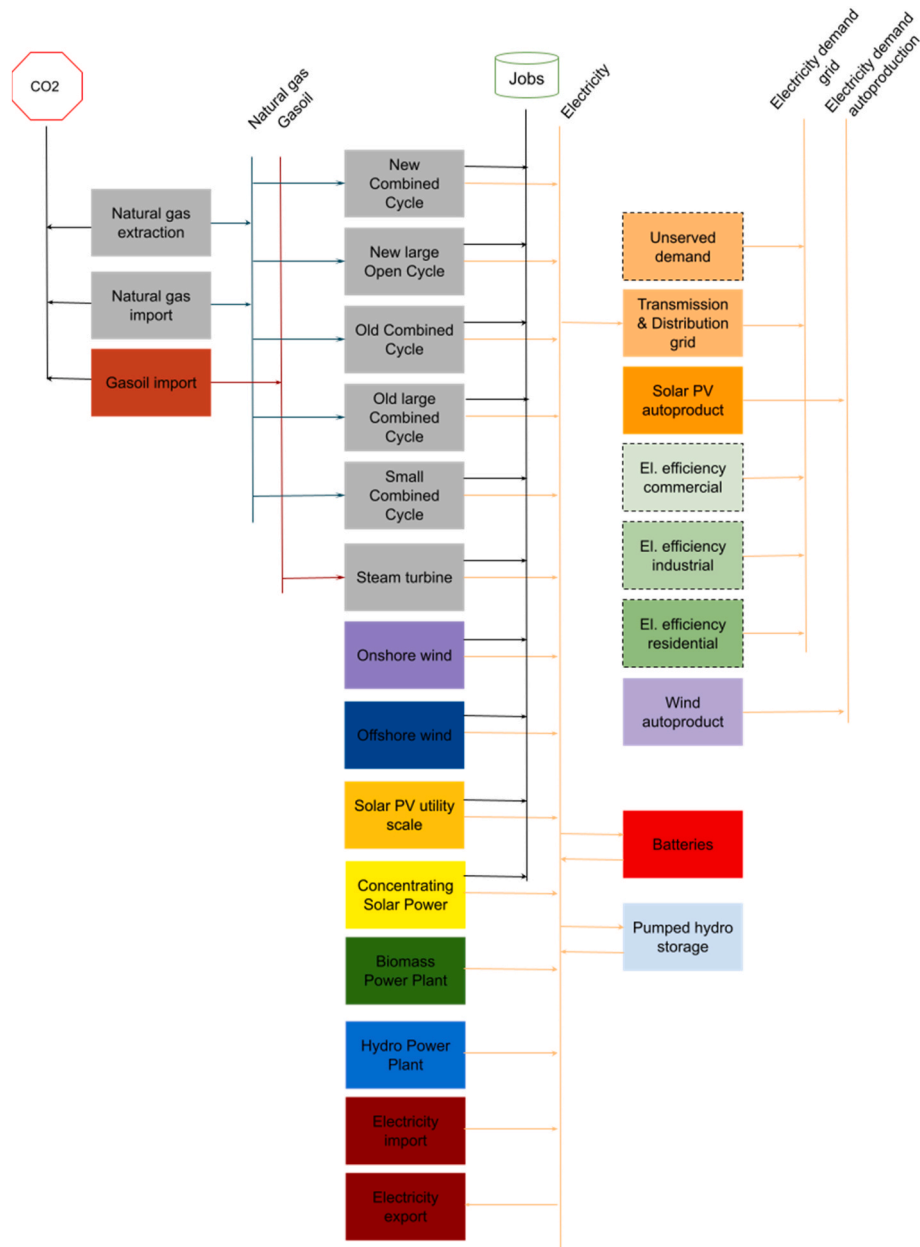


Fig. 1. Schematic representation of the electricity system of Tunisia, as included in the model.

Tunisian Solar Plan (PST) are not changed, both in terms of targets and in terms of technological choices.

- **Lower ambition.** The COVID-19 emergency and the lockdown cause a temporary shift of priorities and consequent delays in the approval of renewable energy and energy efficiency projects in the pipeline. Therefore, it is assessed that the targets of the PST cannot be met by 2030 and the deadline is shifted to 2035. This is represented in the model by shifting the targets for total capacity of wind, solar PV and CSP from 2030 (as in the Constant Ambitions scenario) to 2035. At the same time, interventions to increase energy efficiency in final uses start in 2026, instead of 2021. All other conditions are the same as in the Constant Ambition scenario.
- **Increased ambition.** The government decides to continue the trend of decreased reliance on gas imports forcedly started during the pandemic and avoid the risk of backlashes due to potential new increases of international oil and gas prices. To do so, it encourages more significant and rapid investments in low-carbon electricity supply options and their value chains after the COVID-19 pandemic.

More specifically, it is assumed that the degree of local content of investments in renewable energy technologies and energy efficiency measures is higher throughout the modelling period (not 10–20% as assumed in the Constant Ambition scenario, but rather 40%) and that two new potentially effective technologies are included in the supply mix: offshore wind (in 2028, with speculative assumptions in line with [16]) and grid-connected utility-scale batteries (from 2026). All other policy targets and external conditions are as in the Constant Ambition scenario.

- **Prioritising supply security.** Moving from similar considerations as in the Increased Ambition scenario, after the COVID-19 pandemic the government prioritises investments in energy supply options which may reduce the country's dependence on gas imports. A planned new electricity interconnection with Italy becomes operational in 2027 for imports (600 MW) and in 2031 for exports (500 MW), according to information from STEG. It adds to the existing, and little utilised, interconnection capacities with Algeria and Libya, which amount to 1400 MW in total. In addition, no investments in

Table 2

Techno-economic characteristics of the power supply options.

	Capital Cost	Variable O&M	Fixed O&M	Efficiency	Max capacity factor	Local content	Lifetime
	(MUSD/GW)	(MUSD/PJ)	(MUSD/GW)	%	%	%	Years
New combined cycle	950	0.39	8.52	54%	100%	10%	30
Old combined cycle	–	0.40	8.99	48%	100%	–	30
New gas turbine	642	0.65	10.8	36%	100%	10%	30
Small gas turbine	–	–	20	28.7%	100%	–	30
Hydro	–	–	14.4	–	9.5%	–	100
Pumped Storage	808	–	20	–	–	–	50
Solar PV farm	800↓	–	20↓	–	15% (average)	10%	25
Solar PV autoprod.	880↓	–	22↓	–	15% (average)	10%	25
Concentrating Solar Power (CSP)	4700↓	1.39	141↓	–	45% (constant)	10%	30
Biomass	1250	1.39	62.5	30%	85% (constant)	–	–
Batteries utility scale	83333↓ in MUSD/PJ	0.83 in MUSD/PJ	2500↓ in MUSD/PJ	90%	100%	–	15
Steam Turbine	–	–	21	32.7%	100%	–	40
Wind onshore	1300	2.22	39	–	28.3% (average)	10%	25
Wind offshore	2964	2.22	88.9	–	35.3% (average)	10%	25
Wind autoprod.	1300	–	39	–	28.3% (average)	10%	25
Commercial energy efficiency	2186	–	–	–	–	20%	10
Industrial energy efficiency	2448	–	–	–	–	20%	10
Residential energy efficiency	2623	–	–	–	–	20%	10

Table 3

Electricity demand and gas price projections. Source: STEG.

	2015	2019	2020	2025	2030	2035	2040
Demand (TWh)	15.7	19.8	18.6	24.1	29.5	34.5	39.6
Gas price (USD/toe)	295	316	205	316	387	406	430

expansion of the natural gas import capacity are allowed from 2026 on, meaning that from then on any incremental need for gas is met through import of Liquefied Natural Gas (LNG). The import price of LNG is set according to Pappis et al. [17] and it varies from 9.8 MUSD/PJ in 2015 to 14.2 MUSD/PJ in 2040.

3. Results and discussion

The first insight is that the current and projected change in electricity demand and profiles contingent to the COVID-19 crisis seems not as significant as to affect future energy supply capacity expansion plans to a great extent. Fig. 2 shows the long-term least-cost electricity supply capacity mix under these assumptions, for the Constant Ambition scenario. The trends shown in the figure are representative of those of all scenarios. The supply capacity (Fig. 2a) increases steadily despite the temporary drop in demand. By 2030, the least-cost supply capacity by renewables exceeds the capacity targeted in the PST. Specifically, the capacity of onshore wind turbines just reaches the minimum targets imposed (1755 MW) and does not increase further, showing that the technology is not competitive under the assumptions of this study. The capacity of solar PV at utility scale overshoots the target of 1510 MW in

2030 by a significant amount. It reaches almost 1600 MW by 2023 due to the model assumptions (remember that capacity expansions are constrained by committed plans up to 2025), but then it increases steadily up to almost 3400 MW in 2030 and 6400 MW in 2040. The new capacity of biomass power plants settles on the assumed value of 100 MW.

The capacity of and electricity supply from natural gas remains significant throughout the modelling period. New capacity (beyond the current plans) of combined cycles for baseload generation (cumulatively, around 3670 MW by 2040) and a certain capacity of open cycle gas turbines for peak load generation (cumulatively, 1630 MW by 2040) are installed.

A sensitivity analysis is performed, where the economic recovery is assumed to be slower and, as a consequence, the increase in electricity demand as well: instead of reaching pre-pandemic values right by 2021, the demand is assumed to recover by 2023. This does not bring significant changes in the long-term supply capacity trends, except that investments in renewables are slightly delayed. The PST targets are still overshoot.

The second insight is that the investments prioritising new low-carbon technologies and security of supply are potentially hindered by high upfront capital costs but are competitive in terms of operation costs and use of resources. In the Increased Ambition scenario, off-shore wind results not cost-competitive in terms of capital costs (it would not be installed, if not forced into the model). However, if the investment is assumed as given, the low operation costs and the absence of fuel costs make the project a competitive solution for baseload generation, up to its supply potential (Fig. 3). On the contrary, batteries are not cost-

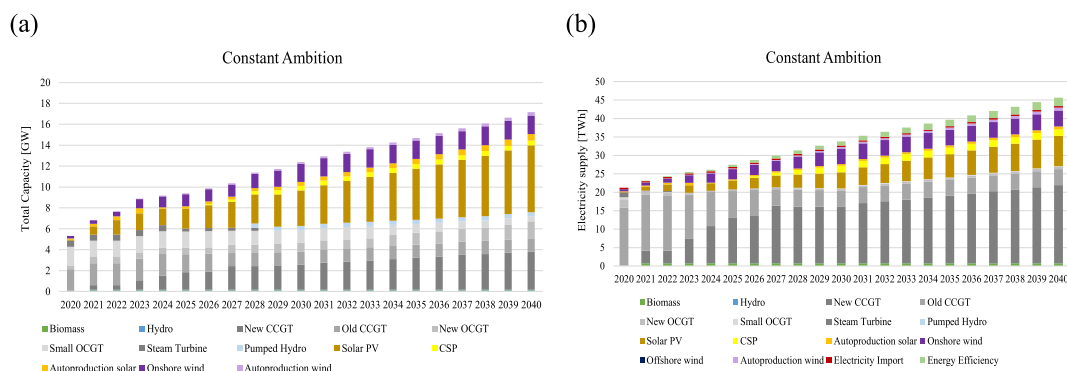


Fig. 2. Installed power supply capacity (2a) and electricity supply (2 b) in the Constant Ambition scenario.

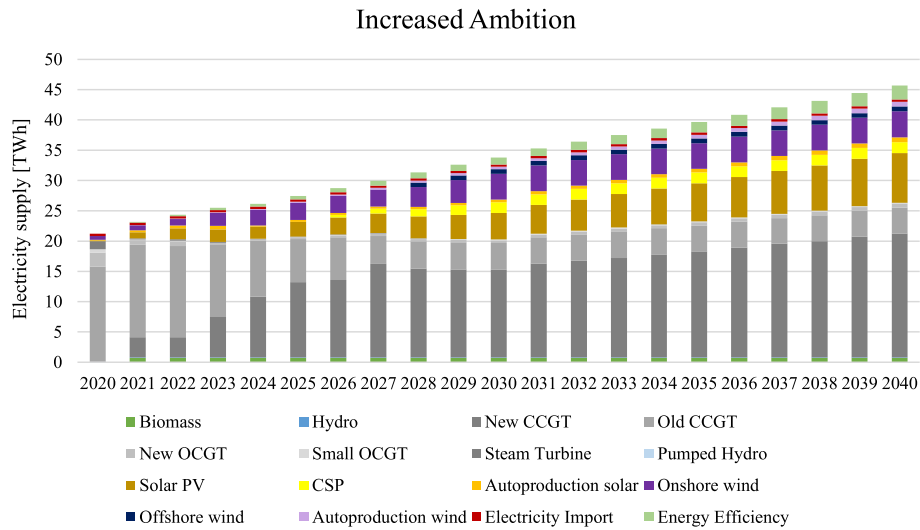


Fig. 3. Electricity supply in the Increased Ambition scenario.

competitive, nor necessary under any condition. Both CCGTs and OCGTs are present with enough capacity to provide the reserve needed with increasing shares of variable renewables. However, the role of batteries and the frequency response of the system in these conditions need to be analysed in more detail with higher resolution models. It is to be noted in Fig. 3 that natural gas accounts for the largest share in electricity generation, in this scenario like in all others. A non-negligible share of the needs is also covered by energy efficiency measures (light green area) and off-grid generation (both through solar panels and wind turbines).

In the *Prioritising Supply Security* scenario, both the new interconnection with Italy and the regasification infrastructure take an important role in the energy supply. Fig. 4 shows electricity imports from Italy to become key in the country's supply as soon as the interconnection becomes operational in 2027.

By running sensitivities around this scenario (varying the prices of electricity export and the availability of the interconnections), we notice that electricity exports become competitive only above a certain sale price threshold (i.e. around 5.4 USD cents/kWh) and considering high availability of the line. If allowed to an extent sufficient to meet the domestic demand, LNG replaces almost entirely the supply of domestic and imported natural gas.

The third insight is that, under the assumptions of this study, the reduction in gas prices caused by the pandemic does not have significant effects on the competitiveness of renewable generation versus gas generation. We performed a sensitivity analysis over the Constant Ambition scenario, assuming more pessimistic projections for gas prices, i.e. halving of gas prices in 2020, and a slower recovery. The halving is meant to reflect projections by the International Energy Agency, which indicate that under the pandemic some oil-indexed gas prices could fall by half over 2020 [18]. The slower recovery assumes that gas prices on average reach pre-pandemic levels only by 2030. This is a fictitious hypothesis, that by the time this paper is reviewed seems not to be verified in reality due to several supply-demand shocks happening worldwide. Nonetheless, it is deemed worth investigating as a scenario that could happen in the future. Under these assumptions, solar PV investments still overshoot the PST targets and vary marginally, compared to the Constant Ambition scenario.

The final insight is that in all scenarios the expected contribution of low-carbon solutions (both investments in renewables and energy efficiency) towards job creation is high. Under the assumptions of the Input-Output model used for this analysis, the effects of the COVID-19 pandemic lead to net job losses in the short-to medium-term across the whole Tunisian economy in all scenarios. This seems to be highly

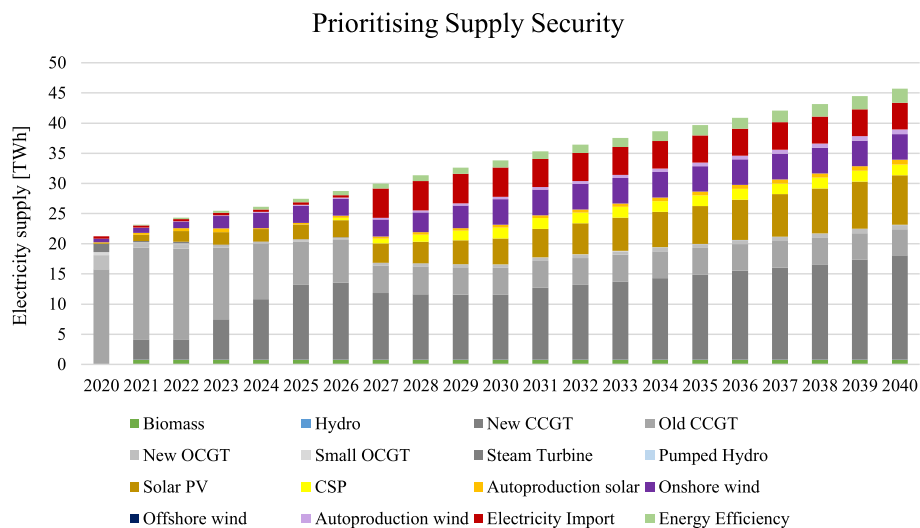


Fig. 4. Electricity supply in the Prioritising Supply Security scenario.

dependent on the local content of the investments in new – gas or renewable – technologies, which is assumed to be on average very low in all cases. The job losses are lowest in the Increased Ambition scenario, which differs substantially from the others in that it assumes higher local content for all new investments. In general, investments in new gas infrastructure would cause net job losses, whereas investments in energy efficiency and renewables would cause net job creation, but not as high as to fully counterbalance the losses.

Interesting insights come from comparing the cumulative difference in jobs created in the period 2020–2040 between the above scenarios and a baseline. We assume as baseline a so-called ‘Frozen Future’ scenario, where only the currently committed investments and decommissioning projects are accounted for. Besides those, any future increase in demand is assumed to be met by increased generation of natural gas fired power plants and all other assumptions are the same as the Constant Ambitions scenario.

In all cases but one, there is net job creation annually, compared to the Frozen Future scenario, due to the higher penetration of renewable technologies and energy efficiency. We show in Fig. 5a cumulative job gains relative to the Frozen Future scenario from 2020 to 2030. Given the limitations of the multiplier approach, these results should be viewed with caution – multipliers only ever show a snapshot of the national economy and may not provide a full picture, causing potential overestimation or underestimation of job creation. Yet, the figures indicate rather robust trends that can be expected with energy technology investments. The difference in cumulative job gains between the Increased Ambition (highest growth scenario) and the Frozen Future scenario is highest and around +63,800, especially due to the assumption of higher local content of investments, compared to the other scenarios. This shows a positive effect of investments in the energy efficiency and renewable energy value chains on job creation. The positive effects are also found in previous studies, even if built upon different model infrastructure and assumptions [19]. The scenario with second highest job creation (and lowest emissions, as per Fig. 5b) is the Prioritising Supply Security scenario. However, for this it must be taken into account that a large share of domestic electricity generation is replaced by imports from 2027 on and this may cause additional job losses not accounted for.

Important insights are gained by complementing the scenario comparison based on the job gains with a comparison based on CO₂ emissions (Fig. 5b). Given the dominance of gas-fired generation, the Frozen Future scenario presents the highest emissions. Most of the other scenarios lie on similar levels of annual emissions and well below the Frozen Future. The Lower Ambitions scenario results in higher emissions until the – delayed – targets are reached in 2035. In Prioritise Supply Security, the emissions decrease sharply from the moment the interconnection with Italy allows imports. It must be noted that this sharp

decrease accounts only for the domestic emissions. The CO₂ content of the imported electricity should be considered for a full picture of the embedded CO₂ emissions of the electricity sector. It is evident that the Increased Ambitions scenario has among the lowest emissions and by far the highest job gains under the assumptions of this study. This highlights the potential combined benefits of investments in energy efficiency and renewable options.

However, almost all the cases (with the exception of the Prioritising Supply Security scenario) show slightly higher emissions by 2040, compared to 2020. The envisaged decarbonisation measures are not sufficient to allow for significant decarbonisation of the electricity sector in the decades to follow. If the electricity demand indeed increases as per the projections, then the share of renewables must be upped significantly and energy efficiency efforts need to be boosted.

4. Policy-relevant insights

In this work we analysed potential long-term implications of the COVID-19 pandemic on the development of the electricity sector of Tunisia and related job creation, with an energy-economy model. We looked at four main scenarios, depicting different ways the future may play out in terms of electricity demand, international fuel prices and policy and investment priorities. The scenarios were co-designed with a consortium of Tunisian institutions and populated mostly with locally available data. The data is openly available (partially aggregated where confidential) and appended to this study. Parametric analyses were run around these scenarios, to consolidate the conclusions and complement them. Although dependent on the sets of assumptions we used, several sets of insights seem to emerge robustly, for the attention of policy makers.

In general, all the least-cost electricity system expansion trajectories analysed in this study overshoot the targets of the Tunisian Solar Plan in terms of installed renewable capacity. The emission trajectories cannot be compared with those of Tunisia’s commitments in the INDCs, since the INDC provides aggregated figures for the entire energy sector and no numerical details on the baseline assumed. The only observation that is worth making is that the INDC shows increasing trajectories of emissions until 2030 for both its baseline and the scenarios with conditional and unconditional commitments. On the contrary, the scenarios in this work show non-increasing emissions during the period 2020–2030, mostly due to the significant investments in renewable technologies proposed by the cost optimisation. Nonetheless, in almost all scenarios the total CO₂ emissions of the electricity sector increase in 2040 compared to 2020. The only exception is the Prioritising Supply Security scenario, only due to the fact that embedded emissions in the imported electricity are not taken into account. This development not only keeps Tunisia far from any deep decarbonisation pathway. It also exposes Tunisia to the

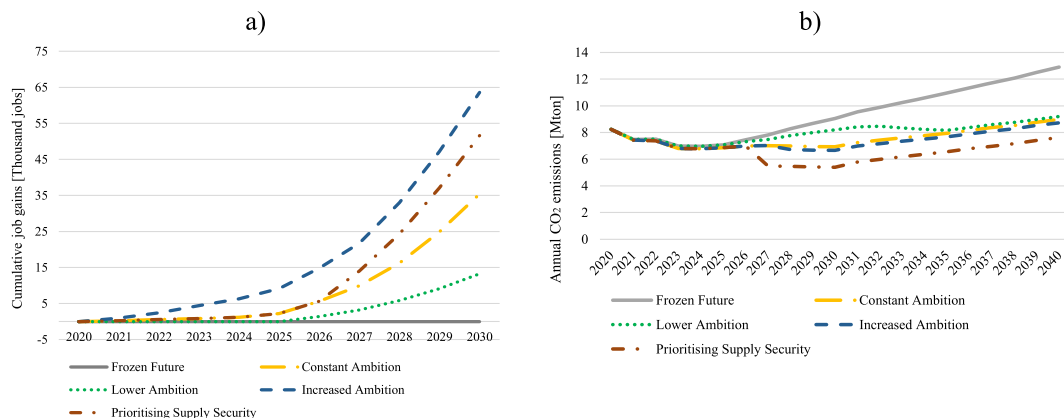


Fig. 5. a) Cumulative job gains by scenario, relative to the Frozen Future scenario; b) Annual CO₂ emissions by scenario.

4. Policy-relevant insights.

risk of heavy economic impacts in case a Carbon Border Adjustment Mechanism is introduced for electricity imports in the EU. To avoid this risk, more ambitious decarbonisation targets should be considered. This argument is reinforced by other insights.

Grid-connected solar photovoltaic emerges as a highly competitive solution in the context and its competitiveness seems not significantly affected by the sharp decreases in gas prices contingent to the COVID-19 pandemic. Its share in the electricity supply could overshoot the PST targets even purely under economic-competitiveness principles. At the same time, the increased reliance on solar photovoltaic and decreased reliance on gas-fired generation, the investments in solar photovoltaic installations and in the photovoltaic production chain could allow significant job gains compared to the business as usual.

Wind turbines emerge as not competitive compared to other options, under the assumptions of this study. However, their cost-competitiveness could change significantly if reductions of capital costs jointly with increases in capacity factors were to be achieved. This may happen, for example, with breakthroughs in offshore wind turbines with floating platforms.

Technically, the gas-fired capacity connected to the grid is expected to slightly increase and to be refurbished. This, and potentially the installation of the connection with Italy, seem to allow the grid to back up the variability of even significant shares of photovoltaic without the need for batteries. Such conclusion needs to be confirmed by power dispatch and grid reliability studies that take as inputs the capacity expansion results of our scenarios. In brief, we suggest that in-depth studies be carried out based on ambitious assumptions on the penetration of renewables and the interconnection with Italy, to confirm the viability of more significant shares of solar than foreseen in previous commitments of the country. However, if the contribution of gas-fired generation remains significant, Tunisia is exposed to the high variability of gas prices (for which remarkable examples came about in recent years). This presents another reason for higher decarbonisation ambitions to be considered.

The contribution of investments in renewables and energy efficiency measures towards job creation is high, though highly dependent on the local content of the investments. This makes an ambitious expansion of renewable energies and energy efficiency and their value chains potentially an attractive option for the design of economic stimulus packages for a green recovery. The coupling with other new value chains, such as the production of green hydrogen for export could strengthen the business case, besides potentially counterbalancing the variability of renewables in the electricity supply. If the country's objectives are to foster decarbonisation while creating jobs, a policy priority may aim to support the local production and construction of renewable technologies. In the case of solar PV particularly, this would also correspond to a highly cost-competitive solution, at assumed costs. Studies dated late 2020 suggest that the local content of investments in renewables may currently lie between 20% and 40% (in the range of the Increased Ambition scenario). Under this scenario, in our work job creation is greatest. If greater parts of the value chains were to be concentrated in Tunisia, the effect could be increased. Mechanisms to incentivize the concentration of the renewables value chains in Tunisia could include support to research and innovation (R&I) on one side and direct support on the other. As far as R&I is concerned, models under development in the EU (such as the partnership towards a competitive European industrial battery value chain [20]) could give inspiration. Collaborations through the African Union – European Union partnership may provide the means to support such R&I efforts. As far as direct support is concerned, different types of subsidies to the local production of components and final products could be studied, for instance in the form of fiscal incentives or loans with reduced interest rates.

On the side of the energy efficiency measures, it may be important to reflect upon which measures should be promoted by local directives, through ad-hoc programmes and incentives for end-users (e.g. tax reductions). Preliminary analyses in the MethodsX papers linked to this

work show that fast payback options may bring in the short term benefits in terms of job creation even if implemented with imported goods.

5. Limitations and future work

The analysis relies on numerous assumptions. On many assumptions around the electricity system investments and the techno-economic characteristics of well-established technologies there is relatively low uncertainty, thanks to the inputs from the local consortium including utilities and government. The demand projections are more uncertain, considering that the pandemic may speed up industrial shifts that ripple through international value chains. Uncontrollable levers such as fossil fuel prices always present the highest uncertainty. This was partially contained by running sensitivity analyses. Also the technological learning of renewables, especially wind and solar PV, has proven to be quite uncertain and often underestimated. This study could be subject to this weakness, which can be addressed in future work in different ways: by updating the assumptions, possible thanks to the availability of the dataset and the open source model infrastructure; and by incorporating technology learning according to methods discussed in literature [21]. On the technology portfolio, aspects requiring immediate attention would be re-assessment of the potential of biomass in the electricity mix and refining of the data base for batteries and other storage solutions.

The assumptions around the economic model are the ones carrying the highest uncertainty, especially those on average wages, wage share of value added and local content. These assumptions highly impact the calculation of the job gains and job losses coefficients. Although the local content was discussed with local economists, its quantification requires itself detailed ad-hoc studies and its uncertainty can probably be reduced only to a certain extent. One important assumption we made regarding the local content is that, when it changes in the Increased Ambition scenario, it does so throughout the model period and not gradually. We made this assumption with the aim to just show the impact of different shares of local content and not represent a realistic progression of it through the years (which is itself very uncertain). Our suggestion for users of the model is that they assess the importance of each assumption by running large sensitivity analyses. We strived to make this possible by delivering an open access dataset, creating open source model infrastructure and keeping the complexity of the energy-economy frameworks manageable.

Finally, we note that the IO model we use is built upon a snapshot in time of the Tunisian economy, with fixed structure and monetary flows between sectors.

All the limitations and potential improvements of the methodology are described in higher detail in the MethodsX papers linked to this work.

Author contributions

Francesco Gardumi: conceptualisation, Methodology, Data curation, visualisation, Investigation, Writing – original draft, revision; Nesrine Mhiri: Data curation, Investigation; Mark Howells: Methodology; Franziska Bock: Data curation, review, editing, revision; Thameur Necibi: Data curation, review; Chiheb Bouden: review.

Data availability

The data supporting this analysis is openly available under license CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>) and annexed to this publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.esr.2021.100759>.

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